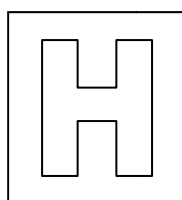


Candidate Name: _____

Class Adm No

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2014 Preliminary Examination II

Pre-university 3

H2 Biology

9648/03

Paper 3

22 September 2014

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
Total	

This question paper consists of 14 printed pages.

[Turn over

Answer **all** questions

- 1** The screwworm fly, *Cochliomyia hominivorax*, is a pest of livestock. The size of its population can be reduced by releasing large numbers of sterile male flies into the environment. The sterile male flies mate with wild female flies, but no offspring are produced.

Large-scale release of genetically modified (GM) sterile males in the 1950s took place in southern USA. Each GM fly contained a gene coding for green fluorescent protein (GFP) taken from a species of jellyfish, used as a genetic marker. The added DNA also included a promoter.

- (a)** Explain the purpose of using a genetic marker in genetic engineering.

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..... [2]

- (b)** Explain why genes for fluorescent proteins such as GFP are now commonly used as genetic markers instead of genes for antibiotic resistance.

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..... [2]

- (c)** Explain why a promoter needs to be included when transferring a gene from jellyfish into the screwworm fly.

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..... [2]

Although GFP is visible in the screwworm fly, allowing detection of GM flies, confirmation of the presence of the newly added gene can be done via genetic fingerprinting. The first stage of genetic fingerprinting involves using restriction enzymes to digest fly genomic DNA and conducting gel electrophoresis.

- (d) Explain how the presence of the GFP gene can be detected after gel electrophoresis is complete.

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..... [3]

Fig. 1.1 shows the restriction map of the plasmid that was used to clone the sterility gene into male screwworm flies.

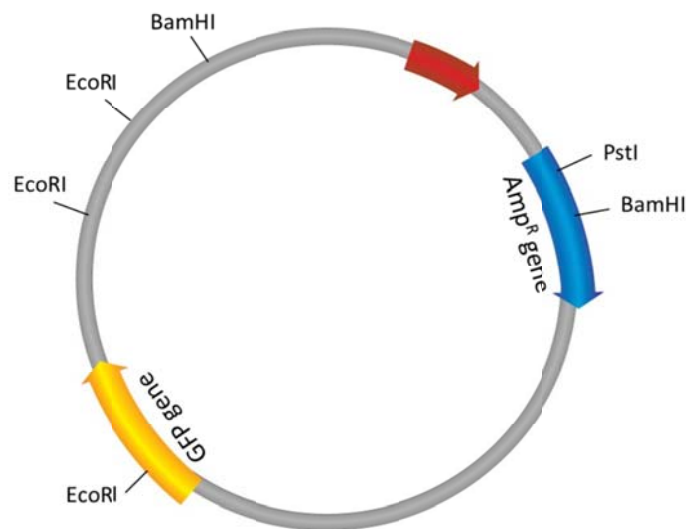


Fig. 1.1

- (e) Suggest and explain which restriction enzyme should be used to cleave the plasmid for insertion of the sterility gene.

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..... [2]

Before the gene for sterility can be cloned into the plasmid, scientists have to generate many copies of the gene using the Polymerase Chain Reaction (PCR). Fig. 1.2 shows the changes in temperature at different stages of the PCR process.

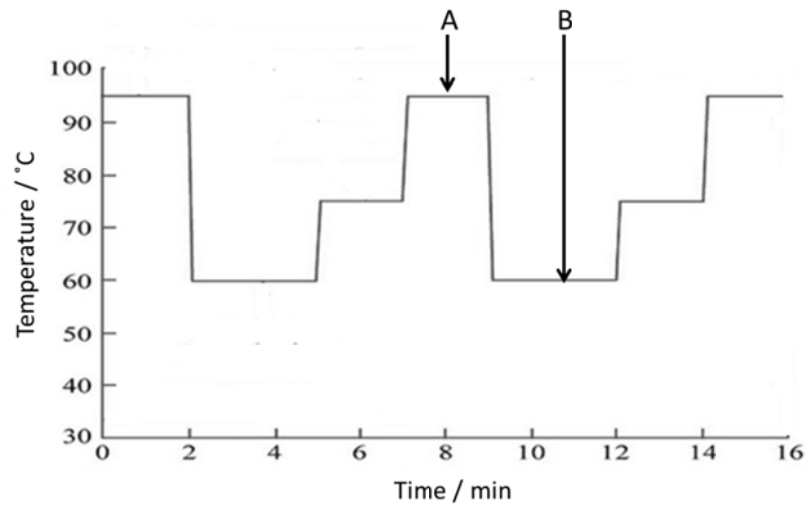


Fig. 1.2

- (f) With reference to Fig. 1.2, explain the purpose of stages A and B.

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..... [4]

- (g) When one scientist conducts PCR using the conditions shown in Fig. 1.2, he finds that the final mixture contains DNA fragments of varying lengths, instead of only one length.

Suggest why the final mixture contains DNA fragments of varying lengths.

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..... [1]

[Total: 16]

- 2 Severe combined immunodeficiency (SCID) is a genetic disorder characterised by the absence of functional T-lymphocytes. There are two common forms of SCID, namely X-linked SCID and adenosine deaminase (ADA) deficiency SCID.

- (a) Explain the difference in the cause of X-linked SCID and ADA deficiency SCID.

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One method of treating ADA-deficiency SCID is through the use of gene therapy, where stem cells are isolated and the non-functional ADA gene is replaced with the functional gene. The optimal number of functioning white blood cells is approximately 5000 – 8000 per mm^3 of blood.

Two patients undergo the same stem cell gene therapy for ADA-deficiency SCID, using stem cells isolated from their own bodies. Retroviral vectors were used to transport the functional ADA gene into isolated stem cells. The results of the therapies are shown in Fig. 2.1.

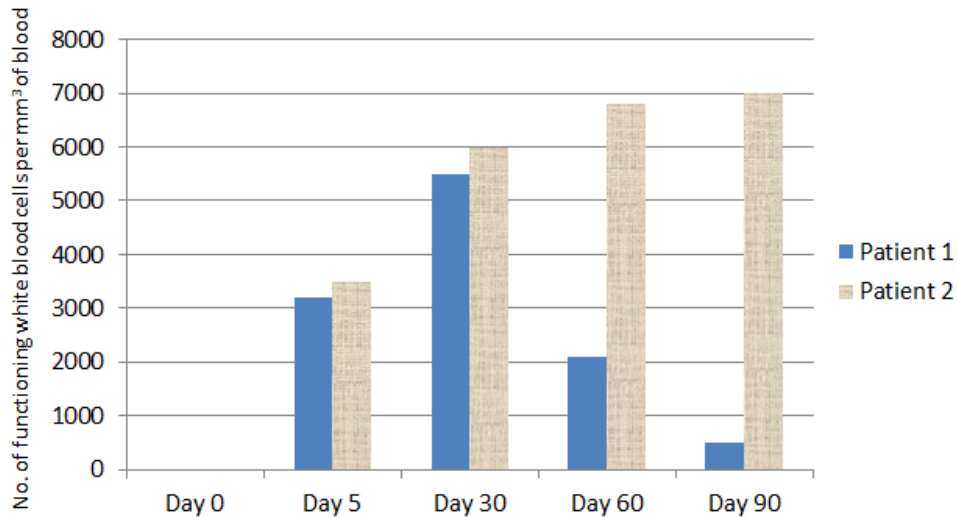


Fig. 2.1

- (b) Suggest and explain what type of stem cells could be used for treatment of SCID.

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..... [2]

(c)

- (i) With reference to Fig. 2.1, describe the differences between the effects of gene therapy in patient 1 and patient 2.

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- (ii) Suggest an explanation for the difference in the results of treatment for patients 1 and 2.

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- (d) Explain why doctors prefer to use stem cells for gene therapy as compared to somatic cells.

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- (e) Suggest two advantages of using viruses to transport the ADA gene into stem cells.

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..... [2]

[Total: 12]

- 3 Duchenne muscular dystrophy (DMD) is a genetically inherited disorder involving a deficiency in the dystrophin protein. Individuals with the disease experience progressive muscle weakness and eventual death. DMD is more common in males than females.

Fig. 3.1 shows a pedigree chart for a family with inheritance of DMD and the restriction pattern of different members of the family obtained by detection of an RFLP closely linked with the DMD gene.

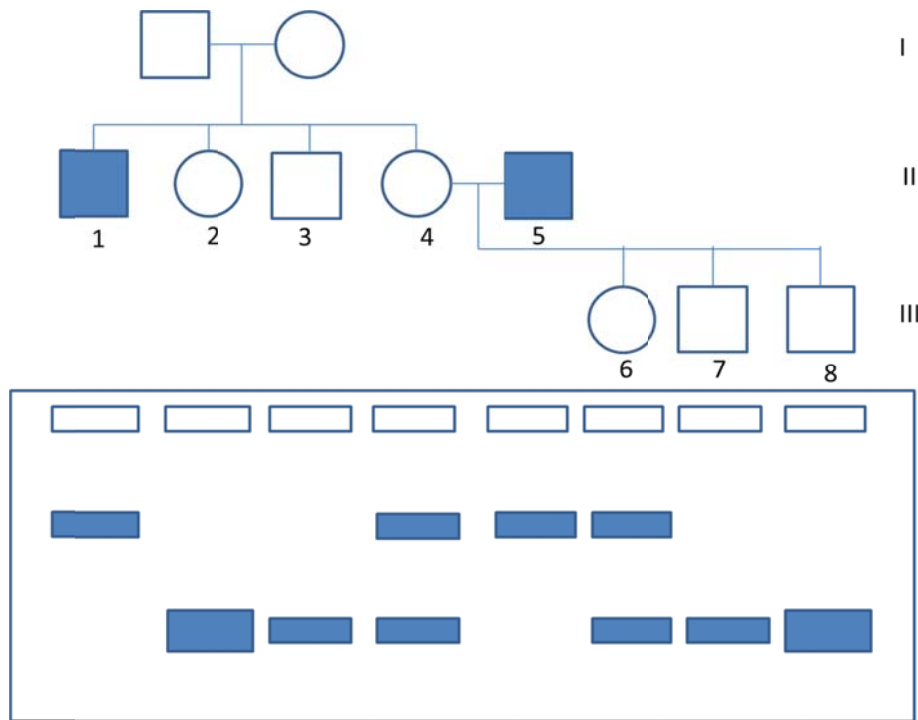


Fig. 3.1

(a)

- (i) With reference to Fig. 3.1, state the mode of inheritance of DMD.

..... [1]

- (ii) Based on the restriction pattern shown in Fig. 3.1, deduce the genotype of individual III-6.

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 [2]

- (iii) Suggest why there is a difference in band thickness between individuals II-2 and II-4 even though they exhibit the same phenotype.

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..... [2]

- (iv) One of the individuals represented in Fig. 3.1 exhibits an anomalous restriction pattern.

Identify which individual exhibits the anomalous pattern and suggest a reason for the anomaly.

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..... [3]

In disease detection, RFLPs are often used as genetic markers to predict whether or not a person is likely to suffer from a particular disease.

- (b) Explain the features that must be present in order for a genetic marker to be used as a means of determining inheritance of a disease.

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..... [2]

- (c) Suggest two other uses for RFLPs other than for disease detection.

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..... [2]

[Total: 12]

4 Planning Question

Bacteria have the ability to acquire new genes via horizontal gene transfer. One of the methods of horizontal gene transfer is the process of transformation, where bacteria are able to take up naked DNA from the surroundings. The process of transformation has been utilised by scientists to introduce foreign DNA into bacteria cells for gene cloning purposes. This is often done using the heat-shock method.

You are required to plan, but not carry out, an investigation into the effect of heat-shock temperature on the efficiency of bacterial transformation, using a range of temperatures between 30°C and 80°C.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you must use:

- Solution of plasmid containing tetracycline resistance gene
- Suspension of *E.coli* cells
- Distilled water
- LB nutrient broth
- Petri dishes containing LB agar medium and tetracycline antibiotic
- Calcium chloride solution
- Micropipettes
- Microfuge tubes
- Stopwatch
- Sterile streaking loops
- Parafilm
- Ice
- Access to hot water at 100°C
- Access to incubator at 37°C
- Alcohol disinfectant

Your plan should have a clear and helpful structure to include:

- an explanation of theory to support your practical procedure
- a description of the method used including the scientific reasoning behind the method
- an explanation of the dependent and independent variables involved
- relevant, clearly labelled diagrams
- how you will record your results and ensure they are as accurate and reliable as possible
- proposed layout of results, tables and graphs with clear headings and labels
- the correct use of technical and scientific terms

[Total: 12]

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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[illegible]

5 Free-response question

Write your answer to the question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate,
- must be set out in sections (a), (b) etc, as indicated in the question.

(a) Explain the advantages of tissue culture techniques over traditional vegetative propagation techniques. [5]

(b) Bovine somatotrophin (BST) is a peptide hormone that is produced by the pituitary glands of cows.

Using your knowledge of genetic engineering, explain how BST could be produced in large quantities using bacteria. [10]

(c) Most genetically modified (GM) crops are only approved as animal feed and not for human consumption.

Suggest why the use of GM crops is not widespread despite its advantages. [5]

[Total: 20]

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